

3-Week Syllabus

Generated: 2026-05-04

Space Mission Design & Operations Programme

3-Week Intensive (Monday-Friday, 9:00-16:00)

Audience: Industry professionals, graduate students, upper-year undergraduates

Capacity: 20-30 participants in teams of 4-5

Tool: SpaceCDF AI-Assisted Concurrent Design Facility

WEEK 1: Canadian Space Landscape & Regulatory Environment

Day	Topic	Activities
Mon	The Canadian Space Ecosystem	Canadian Space Agency mandate & programs. Canadian space industry map (MDA, Telesat, GHGSat)
Tue	Regulatory Framework	ISED spectrum licensing (CPC-2-6-02). RSSSA (Remote Sensing Space Systems Act) for EO missions. Can
Wed	International Standards & Compliance	ECSS standard framework and tailoring for small missions. NASA SEH 17 processes ove
Thu	Mission Needs & Opportunities	Identifying mission needs in the Canadian context (Arctic monitoring, maritime surveillanc
Fri	Space vs Non-Space Trade & Concept	Trade study methodology (criteria, weightings, scoring). Evaluating existing services

WEEK 2: Concurrent Design Facility -- Designing a Mission

Day	Topic	Activities
Mon	System-V & Requirements	System-V model deep dive. SMART requirements from objectives. Functional decomposition. Teams en
Tue	Subsystem Design: Power, AOCS, Thermal	Orbit selection trade. Solar array & battery sizing. Duty cycling. Attitude contr
Wed	Subsystem Design: Comms, Structure, Propulsion	Link budget calculation. Frequency band selection & licensing. CubeSat st
Thu	Integration, Verification, Cost	Interface matrix review. V&V matrix assignment. Environmental test planning. Cost estima
Fri	Design Review & Optimization	Run optimizer. Sensitivity analysis. Gate review (MCR criteria). Resolve conflicts. Generat

WEEK 3: Mission Operations & Simulation

Day	Topic	Activities
Mon	Operations Concept Development	Ground segment architecture. MCS setup (COSMOS/OpenMCT/Yamcs overview). Operations proced
Tue	Mission Operations Training	Console positions and responsibilities. Communication protocols (voice loops, log keeping).
Wed	Mission Simulation Day 1	Full-day simulated LEOP + commissioning. Teams operate their designed mission on console. Deplo
Thu	Mission Simulation Day 2	Full-day simulated nominal operations + contingency. Science data acquisition mode. Ground pass
Fri	Wrap-Up & Presentations	Final design documentation review. Each team presents complete mission: need, design, ops concep

Assessment

Component	Weight	Description
Worksheets (daily)	20%	Completed during sessions, formative feedback
Design Review (Week 2 Fri)	30%	Team presentation of preliminary design
Simulation Performance (Week 3)	20%	Console operations, anomaly response
Final Presentation (Week 3 Fri)	30%	Complete mission design, ops concept, lessons

Prerequisites

- Engineering background (any discipline) or equivalent experience
- Basic familiarity with orbital mechanics concepts (recommended, not required)
- Laptop with modern web browser (Chrome/Firefox/Edge)

Materials Provided

- SpaceCDF tool access (web-based, hosted for the programme)
- Facilitator's Book (reference, post-programme access)
- Learner's Workbook (worksheets, tool guides)
- ECSS/NASA standard excerpts (relevant sections)
- Canadian regulatory filing templates

Outcomes

Participants will leave with:

3-Week Syllabus

- A complete CubeSat mission design (documented, reviewed, ECSS-compliant)
- Hands-on experience with concurrent design methodology
- Understanding of Canadian space regulatory environment
- Mission operations experience including anomaly response
- Network of peers across industry and academia